

RESEARCH METHODS

DAY 1

WELCOME TO RESEARCH METHODS by ResearchCORE!

**Hands-on, fast-paced experience where you will *be* researchers
tackling real-world problems.**

1

Pick a Global Challenge

?

Ask a Question



Gather Information



Analyze the Information



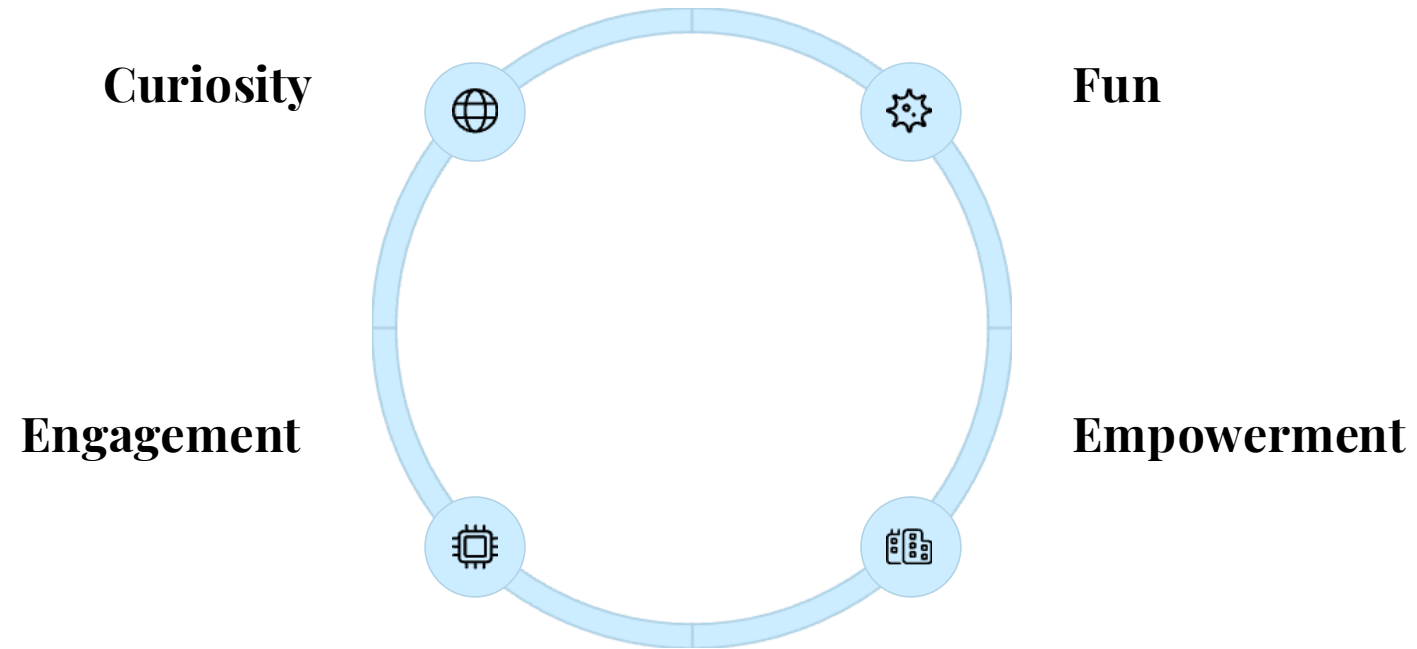
Draw Conclusions



Share Findings - The final deliverables: A formal presentation by Day 10 and a research proposal by Day 9

Journey Ahead: Your Very Own Research Process

Our approach





**Not only to learn how to conduct research,
but to learn how to think like a researcher to**

- Solve problems**
- Challenge assumptions**
- Make smart decisions**

**As you understand how the world works,
you can change it for better!**

Let's Start with a Game

BINGO TIME!

Let's Start with a Game

BINGO TIME!

**Walk around and find people who check the boxes on the bingo card.
Let us know once you check all the boxes in a line.**

B	I	N	G	O
Interested in space	Interested in volcanoes	Loves chemistry	Enjoys robotics	Likes science museums
Has done a science fair project	Enjoys coding	Likes astronomy	Wants to help the environment	Thinks science is fun
Wants to go to Mars	Loves building things		Wants to design apps	Interested in AI
Loves learning about the brain	Is into math challenges	Has built a robot	Wants to study insects	Loves dinosaurs
Wants to be an engineer	Curious about how things work	Watches history videos on YouTube	Interested in understanding human behavior	Wants to explore the ocean

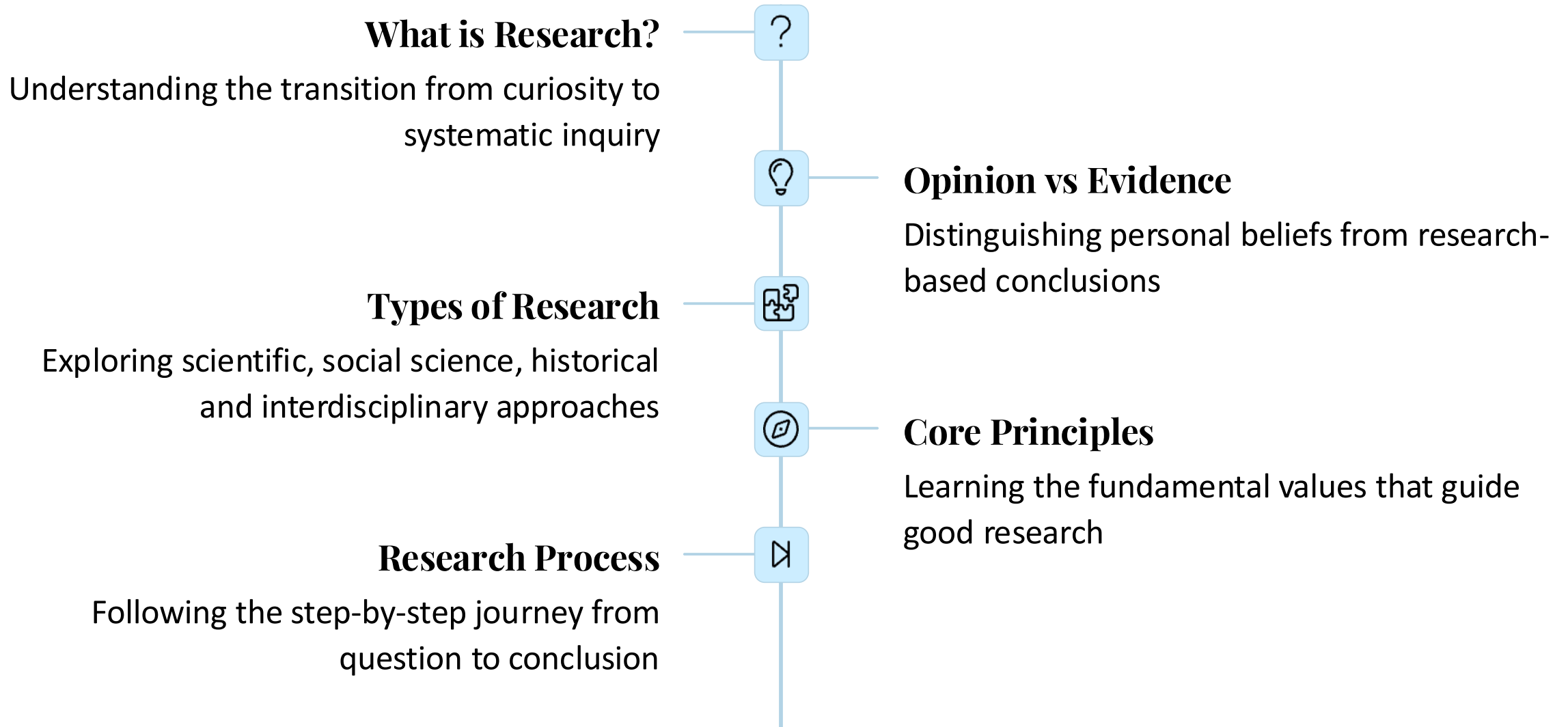
What Is Research?

Let's explore the fundamental concepts of research, distinguish between opinions and evidence-based conclusions, and examine the research process across disciplines.

Michelle Yin & Burhan Ogut



Chapter Roadmap





What is Research?



A Journey of Discovery

Research is a systematic investigation to establish facts and reach new conclusions.



Question-Driven

Every great discovery begins with a well-framed question that guides your direction.



Evidence-Based

Research relies on gathering and analyzing evidence rather than opinions or assumptions.



From Opinion to Evidence: The Schoolday Fatigue Mystery



From Opinion to Evidence: The Schoolday Fatigue Mystery

Eren's Opinion

Students are tired because they stay up too late scrolling through social media. TikTok is ruining sleep habits.

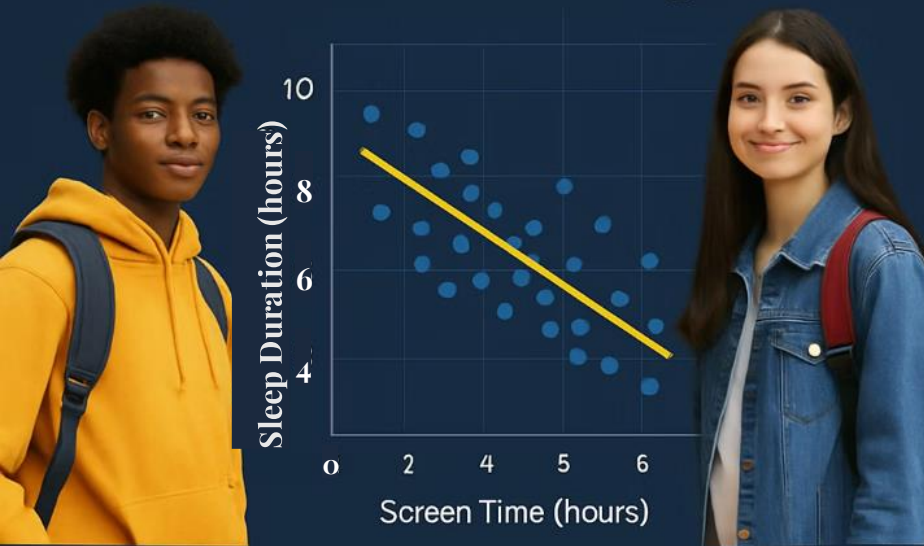
Derin's Hypothesis

Early school start times might be the real issue, not necessarily phone use.

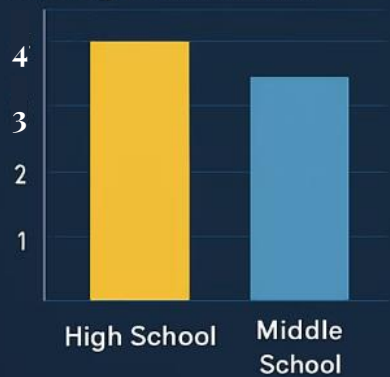
Their Research Approach

Design an anonymous survey, analyze patterns, read studies on teen sleep and blue light effects.

Correlation Between Screen Time and Sleep Duration for Teenagers



Average Screen Time



Average Sleep Duration



High School Middle School

Their Evidence-Based Findings



Nuanced Results

Data showed a more complex picture than either initial assumption.



Phone Correlation

Excessive phone use did correlate with shorter sleep for some students.



Structural Factors

Students with strict screen limits still struggled with early wake-up times.



Multiple Causes

Both personal habits and school schedules contributed to morning fatigue.

From Curiosity to Focused Questions



Curiosity Spark

The urge to know more about something that intrigues you



Wonder

Asking yourself "Why is this happening?" or "How does this work?"



Research Question

Transforming curiosity into a clear, specific question to explore

From Curiosity to Focused Questions



Curiosity Spark

The urge to know more about something that intrigues you



Wonder

Asking yourself "Why is this happening?" or "How does this work?"

WHAT ARE YOU CURIOUS ABOUT?

Transforming Curiosity into Research Questions

1

Pandemic Curiosity

"It's weird that people rushed to buy toilet paper during the pandemic."

2

Research Question

"What factors caused people to panic-buy toilet paper during COVID-19?"

1

Plant Curiosity

"My houseplants grow differently in different rooms of my house."

2

Research Question

"How does sunlight exposure affect the growth rate of my houseplants?"

The Difference Between Opinion and Research

Opinion

What you think or feel about something, based on personal perspective.

- Subjective
- Based on personal belief
- May lack supporting evidence

Research

Systematic investigation that produces evidence-based conclusions.

- Objective
- Based on data and evidence
- Follows methodical processes

Examples of Opinion

1. Everyone should stop using social media completely.
2. It's better to study late at night than early in the morning.
3. Recycling doesn't make much of a difference to the planet.
4. Fast food should be banned in school cafeterias.
5. Climate change is too big of a problem for individuals to make a difference.

Examples of Research Statements

1. **According to the CDC's 2021 Youth Risk Behavior Survey**, which surveyed over 14,000 U.S. high school students, nearly **30% reported feeling persistently sad or hopeless** during the past year.
2. **A 2020 global survey by Pew Research Center involving more than 13,000 adults in 13 countries found that 73% saw climate change as a major threat to their country.**
3. **A randomized controlled sleep study at UC Berkeley** found that **teens who slept at least 8 hours** performed significantly better on memory tasks than those with less sleep. The study involved **105 adolescents aged 13–17**. (*Lopez et al., Sleep Research Journal, 2018*)
4. The **World Health Organization**, based on data from dozens of national health agencies, estimates that **air pollution causes 7 million premature deaths worldwide each year**. (*WHO, 2021*)
5. In a Common Sense Media survey of 1,306 teens aged 13–17, 25% reported that social media had a mostly negative impact on their mental health. (Common Sense Media, 2021)

Let's Play a Game:

OPINION or RESEARCH?

A Stanford University longitudinal study (2019) followed 200 middle school students over two years and found that those practicing daily mindfulness showed lower stress and anxiety levels.

Social media is ruining real-life friendships.

**The 2023 IPCC report, drawing from thousands of peer-reviewed climate studies and global temperature data, concluded with high confidence that human activities are the dominant cause of recent climate change.
(Intergovernmental Panel on Climate Change, 2023)**

You can't be truly successful without going to college.

Homework should be optional for high school students.

A JAMA Pediatrics study (2019) analyzed data from over 6,000 U.S. children aged 8–11 and found that those with more than 2 hours of recreational screen time daily had lower language and thinking skills.

In the APA's 2023 "Stress in America" survey, conducted with 3,185 teens and adults, over 70% of teenagers said they felt overwhelmed by responsibilities in the past month.

Reading printed books is more enjoyable than reading on a screen.

According to a National Sleep Foundation (2020) poll of 1,006 U.S. teens, 87% of high schoolers get less than the recommended 8–10 hours of sleep on school nights.

A Key Attitude of a Researcher: Inquisition

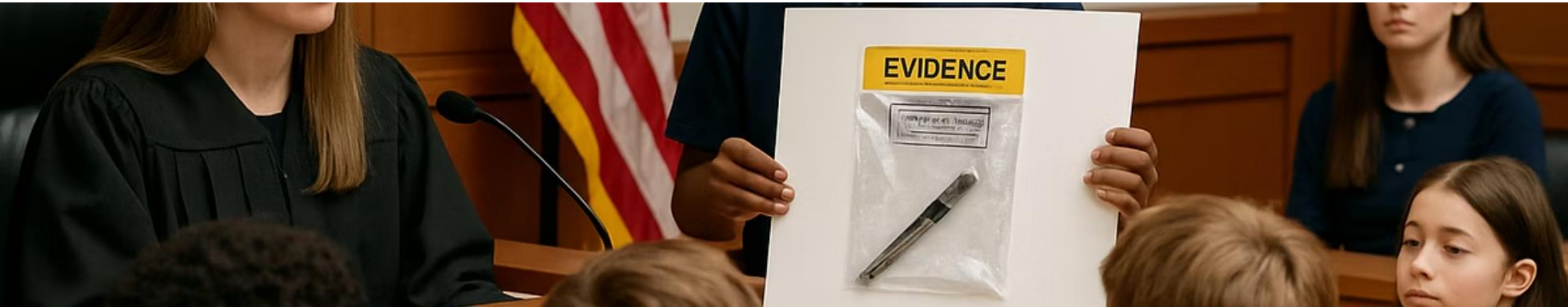
How do I know that?

What information is this opinion based on?

What's that claim based on?

Has anyone actually tested it?

Where is the evidence?



Why Evidence Matters

47%

Misinformation

Percentage of online content containing
inaccurate claims

73%

Better Decisions

Improved outcomes when using
evidence-based approaches

5X

Problem-Solving

More effective solutions when based on
research



Questioning Claims: The Mozart Effect

⁶⁶₉₉



The Claim

"Listening to classical music while studying guarantees an A on the test."

The Inquiry

"What's that claim based on? Has anyone actually tested it?"

The Research

Find scientific studies or conduct a small experiment comparing study methods.

The Evidence

Music may help mood or focus for some people, but doesn't "guarantee" success.

Types of Research: A Multidisciplinary Approach

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Scientific Research

Explores the natural world through experiments and systematic observation

Interdisciplinary Research

Combines multiple fields to address complex questions holistically



Social Science Research

Examines human behavior and social systems through surveys and analyses

Historical Research

Investigates the past through primary sources and contextual interpretation

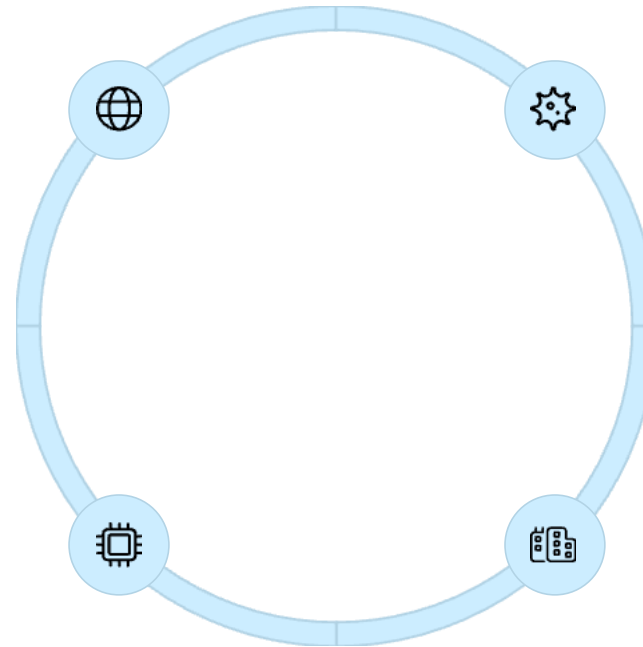
Interdisciplinary Research

Climate Change Research

Combines natural science, economics, sociology, and policy studies

Technology Ethics

Bridges computer science, philosophy, psychology, and law



Public Health Crisis

Integrates biology, medicine, psychology, and historical precedent

Urban Planning

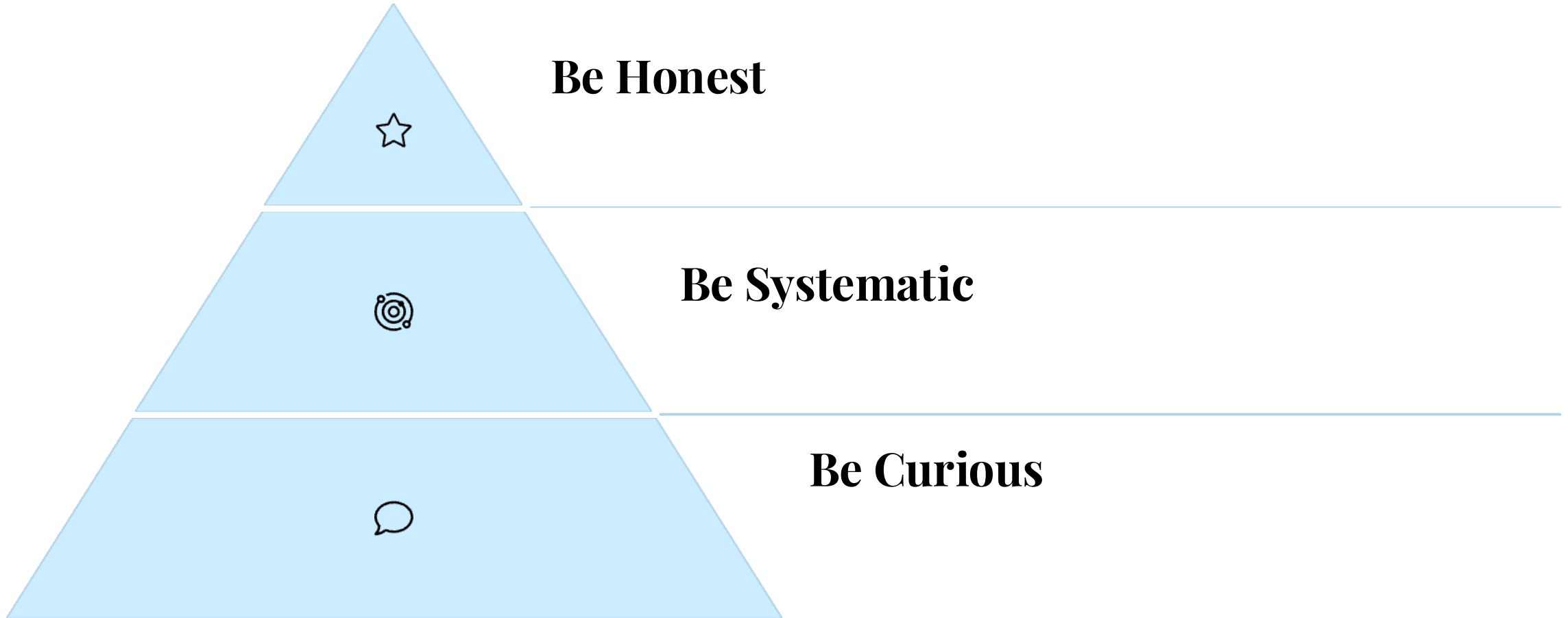
Merges architecture, sociology, economics, and environmental science

Core Principles of Good Research

No matter what the field is

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No matter what the field is



Principle 1: Be Curious

The Starting Point

Curiosity is where all research begins—the spark that ignites investigation.

Everyday Questions

Major discoveries often begin with simple observations or moments of wonder.

Embracing the Unknown

Being comfortable asking "Why?" and "What if?" drives meaningful research.



Principle 2: Be Systematic

Beyond Hunches

Systematic research follows clear procedures rather than random exploration.

Replicability

Others should be able to follow your process and reach similar conclusions.

Organization

Keeping careful records and following methodical steps builds credibility.



Principle 3: Be Honest



Report All Findings

Include inconvenient results that don't match your expectations



Accept Uncertainty

Acknowledge when the data doesn't show clear answers



Give Credit

Cite sources properly and acknowledge others' contributions



Build Trust

Integrity makes your research valuable in school and beyond

The Value of Being Wrong

Embrace Unexpected Results

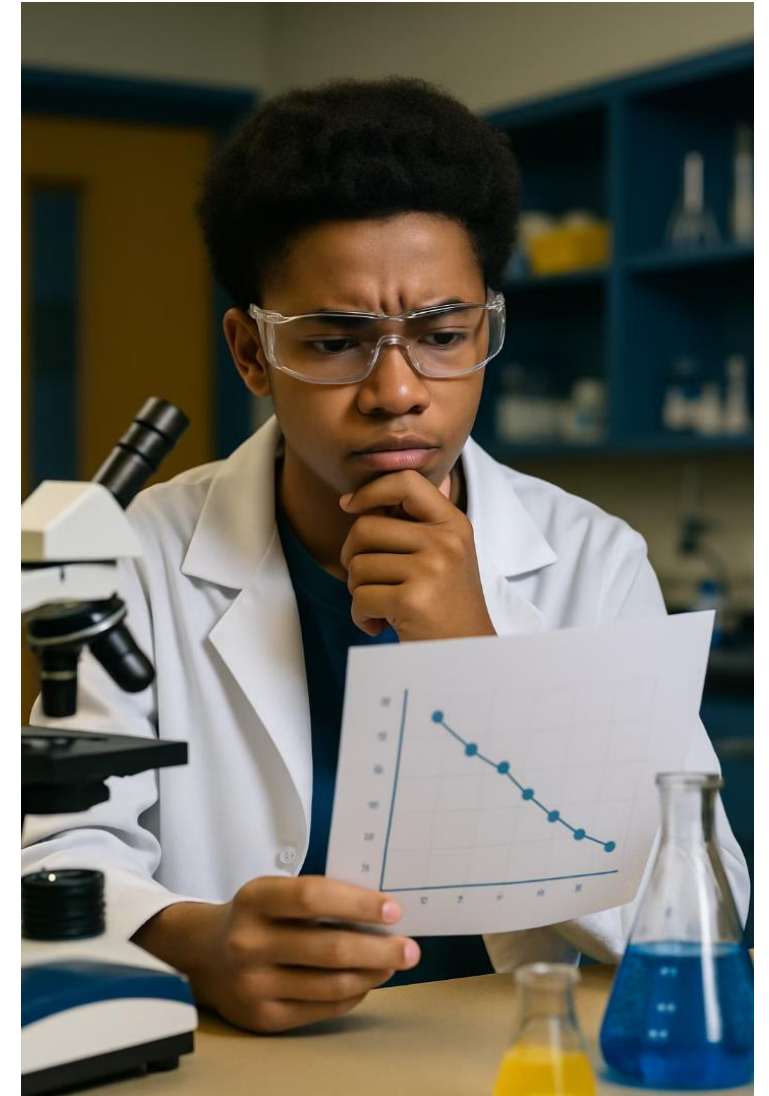
Surprising findings often lead to the most important discoveries in research.

Learning Opportunity

When your hypothesis is wrong, you've learned something valuable about what isn't true.

Intellectual Honesty

Acknowledging incorrect assumptions demonstrates scientific integrity and builds credibility.



The Research Process

The journey from question to insight follows these essential steps:



Ask a Question

Formulate a clear, specific question worth investigating



Gather Information

Collect relevant data through appropriate methods



Analyze the Information

Examine data for patterns, trends, and meaning



Draw Conclusions

Develop evidence-based answers to your original question



Share Findings

Communicate results effectively to appropriate audiences

Real Research Example: AI Art in Education



1

Question

Could AI tools help more students feel empowered to create art?

2

Information Gathering

Surveys, teacher interviews, examples from other schools, literature review

3

Analysis

30% felt confident in drawing skills; 80% would try AI art tools

4

Conclusion & Action

AI art competition could broaden creative access; school launched pilot program

Case Study: School Lunch Research

The Question

Layla wondered: Are students getting nutritious lunches that support focus and learning?

Hypothesis

Students avoid lunches due to taste, freshness, or variety issues—not lack of hunger.

Research Design

- Quantitative: Lunch program participation data
- Qualitative: Student surveys and focus groups
- Observational: Cafeteria behavior patterns
- Stakeholder: Staff interviews



Lunch Research: Findings and Impact



Key Findings

Students wanted fresher options, menu variety, and customization



Solutions Proposed

Student advisory panel, taste tests, nutritional labeling, diverse options



Results

School piloted ideas; lunch participation and satisfaction increased



Lesson

Research can solve real problems and improve everyday experiences

Summary: Key Takeaways



Research Definition

Systematic investigation to answer questions using evidence



Opinion vs. Evidence

Research moves beyond personal beliefs to factual conclusions



Research Types

Scientific, social science, historical, and interdisciplinary approaches



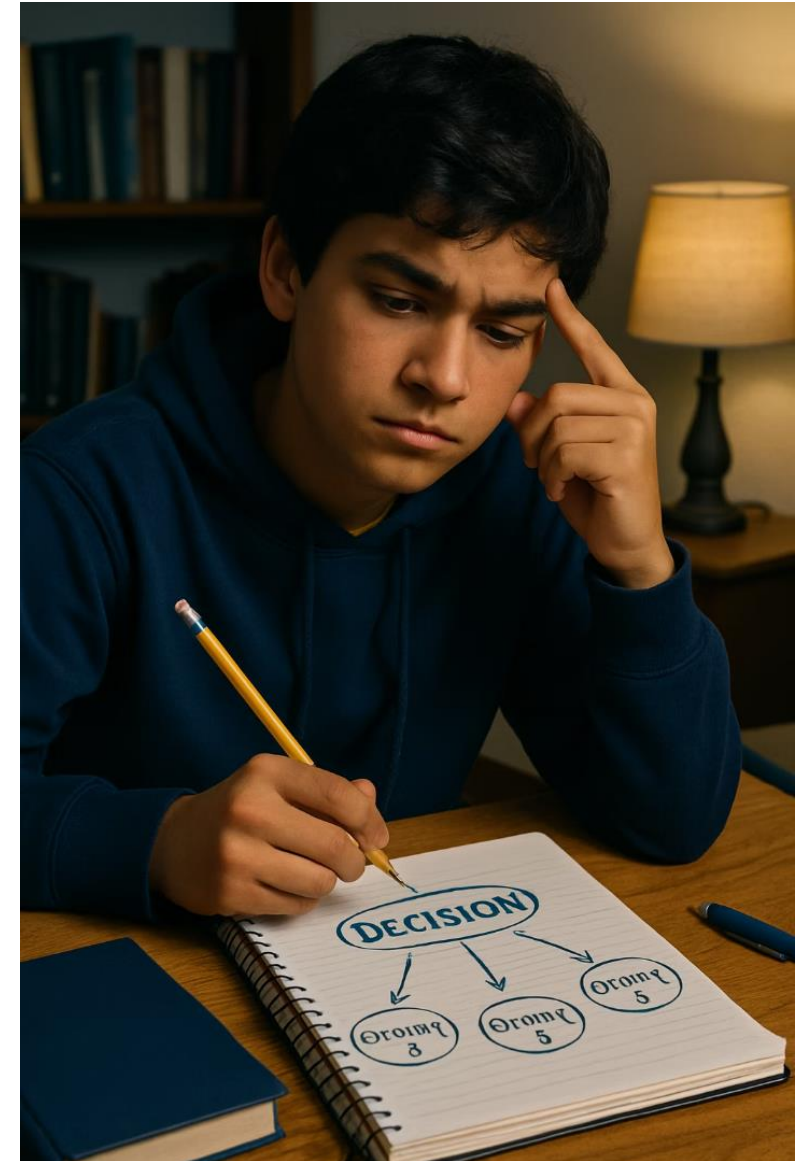
Research Process

Question, gather, analyze, conclude, and share findings

Activity: Find the Research in Your Life

Identify A Recent Decisions

List questions or choices you've faced lately, small or significant



Activity: Find the Research in Your Life

Identify A Recent Decisions

List questions or choices you've faced lately, small or significant

Map Your Research Steps

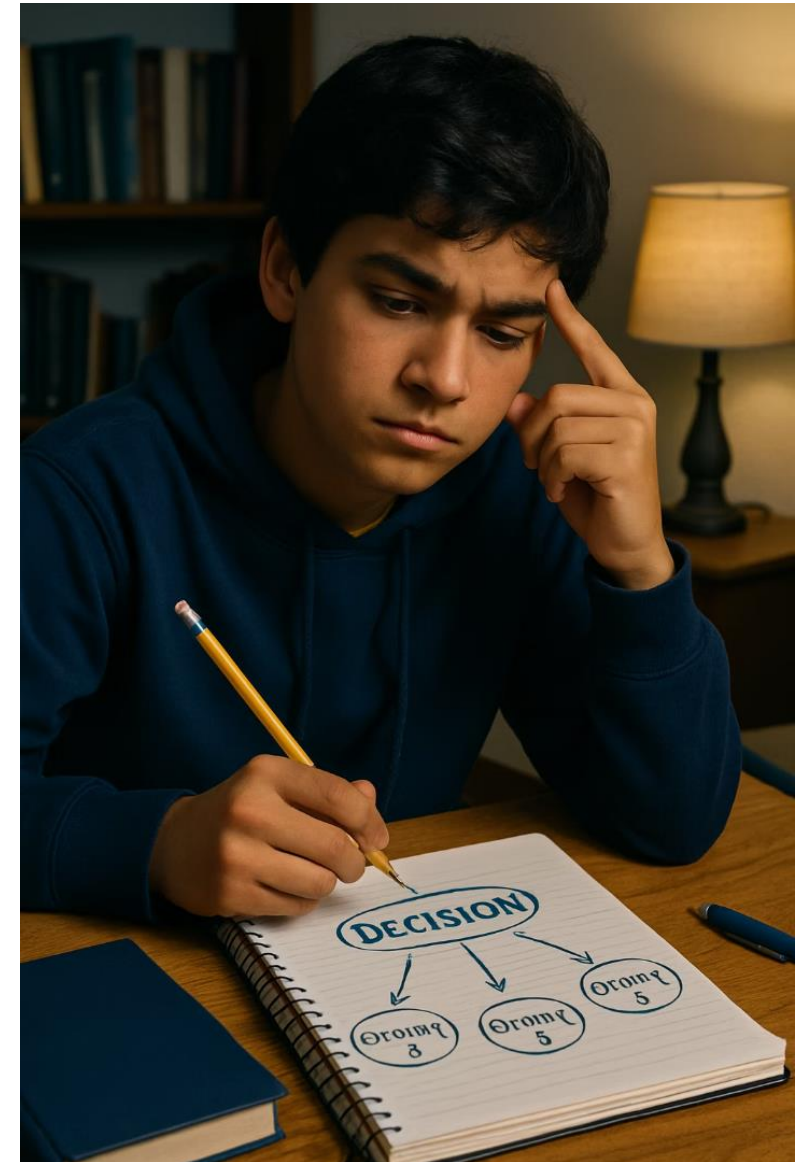
Note how you gathered information before deciding

Evaluate Outcomes

Record your decisions and whether research improved results

Share and Reflect

Compare approaches with classmates or consider your research patterns



GLOBAL CHALLENGES & TEAM FORMATION

1

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?

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Share Findings - The final deliverables: A formal presentation by Day 10 and a research proposal by Day 9

Journey Ahead: Your Very Own Research Process

SIX GLOBAL CHALLENGES

1. Climate Change
2. Renewable Energy
3. Public Health Pandemics
4. Poverty and Inequality
5. Educational Access
6. Digital Privacy

Climate Change

What it's about:

The Earth's temperature is rising due to human activities, mainly burning fossil fuels, which leads to extreme weather, rising sea levels, and ecosystem disruptions.

Key Facts:

- The global average temperature has already increased by **1.1°C (2°F)** since the late 1800s (NASA, 2024).
- **2023 was the hottest year on record**, with ocean temperatures hitting record highs (Copernicus Climate Change Service, 2024).
- By 2050, **over 1.2 billion people could be displaced** by climate-related hazards like flooding, extreme heat, and droughts (Institute for Economics and Peace, 2020).

Renewable Energy

What it's about:

To fight climate change and reduce pollution, we need to shift from fossil fuels to cleaner energy sources like solar, wind, and hydro.

Key Facts:

- In 2023, **30% of global electricity** came from renewable sources (IEA).
- Solar power is now **cheaper than coal** in many countries (BloombergNEF, 2023).
- In 2023, China added **more solar capacity than the rest of the world combined**, installing over 230 gigawatts - more than enough to power all of France (International Energy Agency, 2024).

Public Health Pandemics

What it's about:

Diseases that spread quickly across populations, like COVID-19, can disrupt societies, economies, and healthcare systems on a global scale.

Key Facts:

- COVID-19 led to over **770 million confirmed cases** globally by 2024 (WHO).
- Global vaccine distribution exposed **serious inequalities**: only 26% of people in low-income countries received at least one vaccine dose by late 2022 (Our World in Data).
- In early 2020, China built two emergency hospitals in Wuhan **in under 10 days**, showcasing high-speed infrastructure deployment during health crises (BBC News, 2020).

Poverty and Inequality

What it's about:

Not everyone has equal access to basic needs like food, shelter, education, and opportunity; this gap limits both individual lives and entire communities.

Key Facts:

- Over **700 million people**—about 1 in 11 globally—live on **less than \$2.15 a day** (World Bank, 2024).
- The **richest 1%** of the world's population owns almost **half of all global wealth** (Oxfam, 2023).
- China's poverty rate fell from **88% in 1981 to less than 1% in 2020**, largely through rural development and targeted subsidies (World Bank, 2021).

Educational Access

What it's about:

Millions of children and teens still don't have access to quality education.

Key Facts:

- **244 million children and youth** are out of school worldwide (UNESCO, 2023).
- Girls are **more likely to be excluded** from school in low-income countries, especially during crises like war or pandemics.
- In Sub-Saharan Africa, 9 out of 10 children are unable to read a simple sentence by age 10 due to poor schooling conditions (UNESCO, 2022).

Digital Privacy

What it's about:

As more of our lives happen online, protecting our personal data from being misused by companies, hackers, or even governments is increasingly urgent.

Key Facts:

- In 2023, **over 7 billion data records** were exposed due to breaches (IT Governance).
- Teens are among **the most targeted** age groups for scams and tracking via social media and apps (Pew Research, 2023). Yet, a global survey found that over 60% of teens do not understand how their online data is collected, shared or used (Harvard Kennedy School, 2023).

Looking Ahead: Tomorrow

Formulating Research Questions

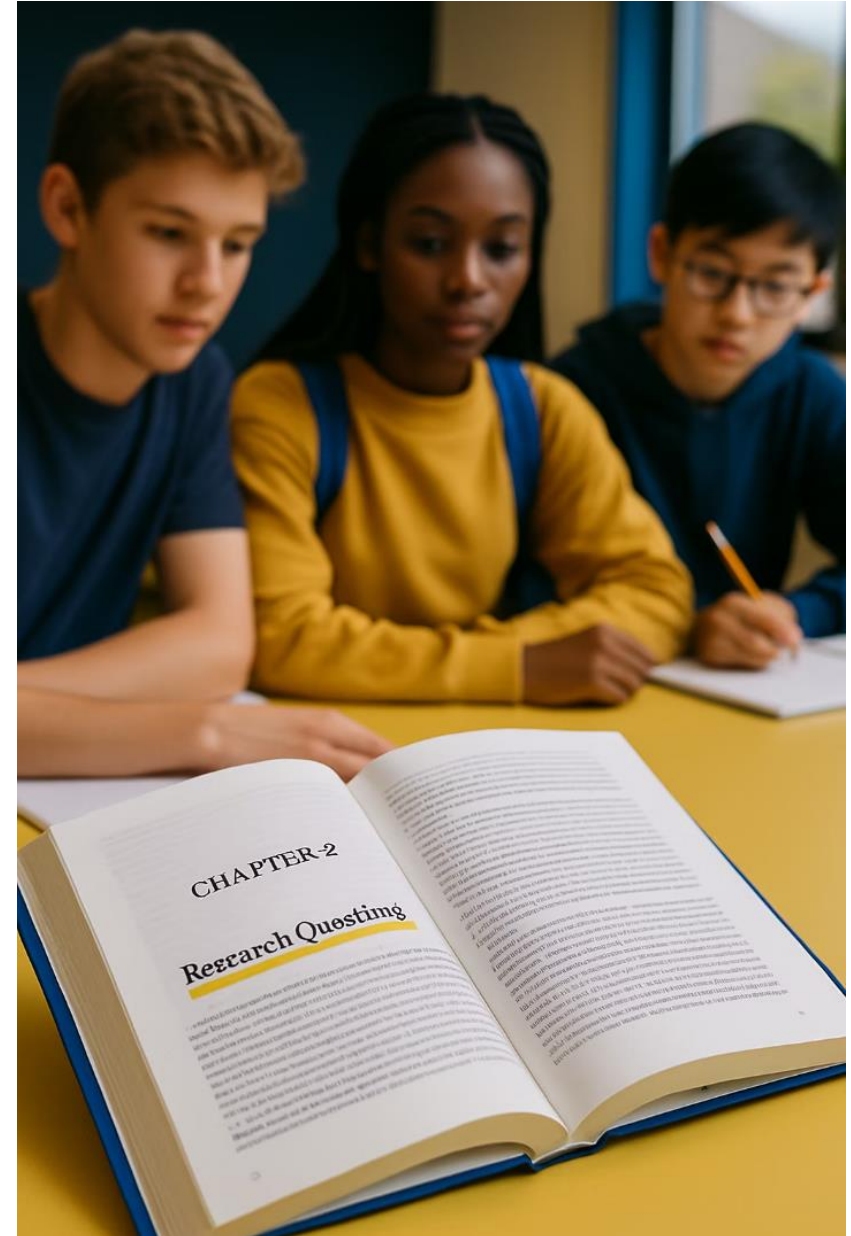
Techniques for developing focused, meaningful questions that drive effective research.

From Broad to Specific

Methods to narrow general interests into precise, answerable research questions.

Evaluating Question Quality

Criteria for assessing whether questions will lead to valuable research outcomes.



Homework: Handout 1

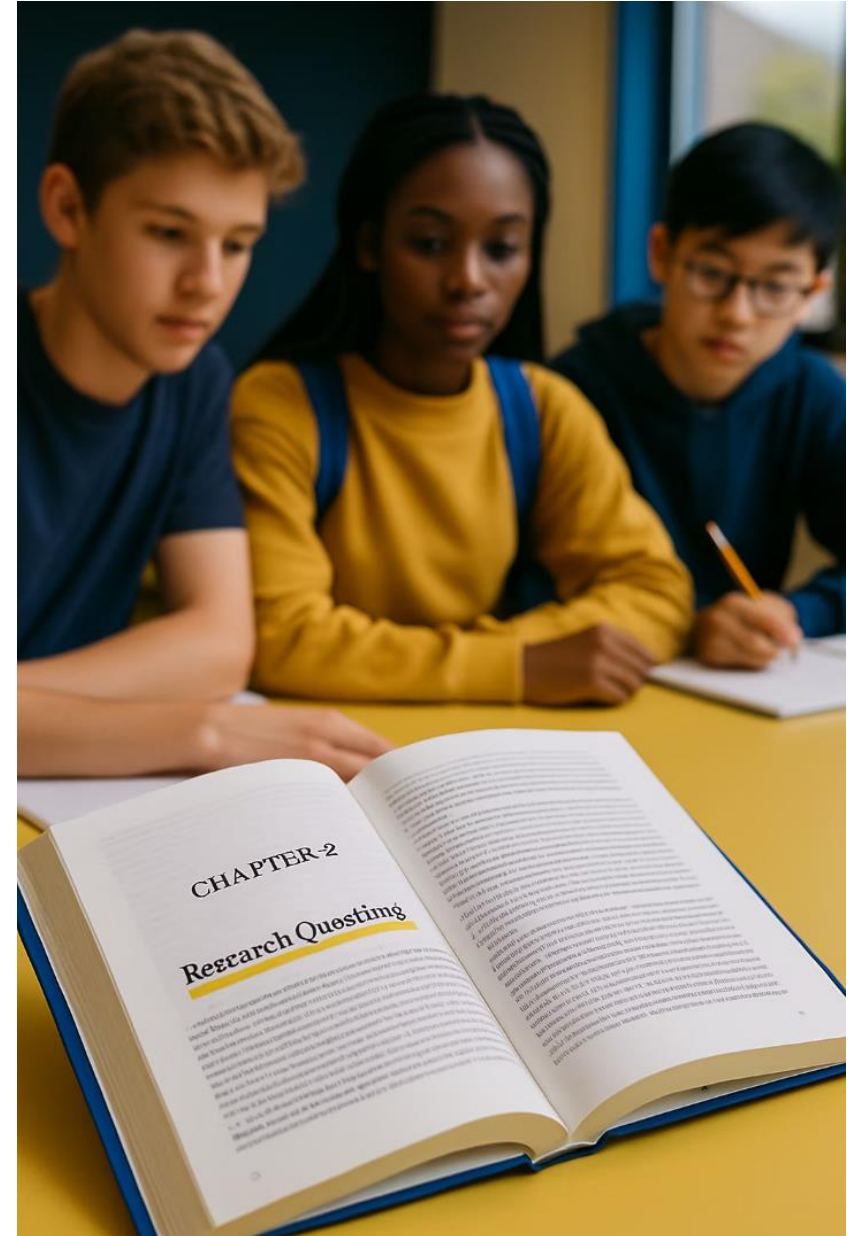
Introduction to Research

Finalize teams

Explore data-sets

Decide on your “Global Challenge”

- Climate Change
- Public Health Pandemics
- Poverty and Inequality
- Renewable Energy
- Educational Access
- Digital Privacy





Let's wrap up this session!

Questions? Final thoughts?

EXTRA SLIDES FOR DAY 1

(DIDN'T HAVE TIME TO COVER)

Let's Play a Game:

FACT (research-supported)
or **MYTH** (misconception)

FACT (research-supported) or **MYTH** (misconception)

“The current global warming trend is primarily caused by natural climate cycles, not human activity.”

✗ MYTH - Research shows over **97% of climate scientists** agree that human activities—mainly burning fossil fuels—are the primary cause of current global warming. (*NASA, 2024 – climate.nasa.gov/evidence/*)

FACT (research-supported) or **MYTH** (misconception)

“As of 2024, more than half of the global population lives in urban areas rather than rural ones.”

 **FACT - Over 56% of people worldwide live in urban settings, a number projected to reach 70% by 2050.**

***(United Nations, 2024 –
un.org/development/desa/publications/)***

FACT (research-supported) or **MYTH** (misconception)

“Studies show that renewable sources like wind and solar could theoretically meet 100% of global energy needs.”

 **FACT - Research suggests a full transition to renewables is technically feasible, though it would require major investment and infrastructure changes. (Jacobson et al., 2017 – Joule, 1(1), 108–121 – sciencedirect.com)**

FACT (research-supported) or **MYTH** (misconception)

“Mental health problems are mostly caused by personal weakness or a lack of willpower.”

✗ MYTH - Mental health conditions are caused by a mix of biological, psychological, and environmental factors, not personal failure. Willpower alone cannot prevent or cure mental illness. (National Institute of Mental Health, 2024 – nimh.nih.gov/health/publications)

FACT (research-supported) or **MYTH** (misconception)

“Most people without access to safely managed drinking water live in sub-Saharan Africa.”

 **FACT - Nearly 400 million people in sub-Saharan Africa lack safely managed water, making it the most affected region. (WHO/UNICEF Joint Monitoring Programme, 2023 – washdata.org)**

FACT (research-supported) or **MYTH** (misconception)

“Biodiversity loss only affects wildlife and has little impact on human life.”

✗ MYTH - Biodiversity loss disrupts food systems, disease regulation, and clean air/water, directly impacting human survival. (IPBES Global Assessment Report, 2019 – ipbes.net/global-assessment)

FACT (research-supported) or **MYTH** (misconception)

“AI makes fair decisions because it doesn't have feelings or emotions like humans.”

✗ MYTH - AI systems can **reproduce bias** if the data or design reflects human inequality. Lack of emotion doesn't equal fairness. (MIT Media Lab, 2019 – gender-shades.org; Nature, 2020)

FACT (research-supported) or **MYTH** (misconception)

“Over half the world’s school-aged children are out of school.”

✗ MYTH - In reality, about 244 million children and youth are out of school—roughly 12%, not over 50%. (UNESCO, 2023 – uis.unesco.org/en/news/244-million-children)

FACT (research-supported) or **MYTH** (misconception)

“Vaccines can help stop pandemics, but only if a large percentage of the population receives them.”

 **FACT - High vaccination coverage is needed for herd immunity to protect both individuals and communities. (World Health Organization, 2022 – [who.int/news-room/questions-and-answers/item/herd-immunity](https://www.who.int/news-room/questions-and-answers/item/herd-immunity))**

NEXT:

GOAL SETTING

WHAT WE KNOW ABOUT THIS ISSUE	WHAT WE WANT TO KNOW	?

WHAT ABOUT YOUR GOALS FOR THE CAMP?

Each team write down 2 goals.